

The Anti-Transfer Paradox: Signal Composition and Markov Structure in Neural Differential Cryptanalysis

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Abstract. Neural differential distinguishers surpass classical methods on round-reduced block ciphers, yet two foundational questions remain open: *where* does the distinguishing signal reside, and does it *compose* across rounds as classical Markov analysis assumes? We answer both through controlled cross-round transfer experiments on three cipher families—SPECK32/64 (ARX), SIMON32/64 (Feistel), and PRESENT-64/80 (SPN)—spanning seven architectures. The signal resides in XOR-difference bits, is architecture-independent (≤ 2.6 pp¹ across seven models), and persists to family-specific depths (SIMON32/64 9r $\gg$ SPECK32/64 7r $>$ PRESENT-64/80 6r). SPN features compose hierarchically ($\geq 96.9\%$ positive transfer), but ARX and Feistel features *anti-compose*: a SPECK32/64 5-round model scores 45.5% on 3-round data ($p < 10^{-7}$), and cross-cipher transfer from SPECK32/64 to SIMON32/64 anti-transfers at 48.9% ($p = 0.004$). We formalize this as the *anti-transfer paradox*: the network extracts 93.8% of native-model mutual information from lower-round data yet maps it to the wrong class. Conditional MI estimation ($\leq 4.5 \times 10^{-5}$ bits) and VAE-based Markov tests confirm the cipher is memoryless—the anti-composition is a property of the neural representation, not the cipher.

Keywords: Neural cryptanalysis · Differential distinguishers · Markov property · Transfer learning · Lightweight ciphers

1 Introduction

Lightweight block ciphers—built on ARX (Add-Rotate-XOR), Feistel, or Substitution-Permutation Network (SPN) paradigms—protect billions of IoT sensors and embedded controllers [9,4]. Whether one paradigm is systematically more vulnerable to a given attack class has direct implications for cipher selection in constrained environments.

Classical differential cryptanalysis [8] answers this through exhaustive search over differential trails, an approach that becomes intractable as round counts grow. In 2019, Gohr [12] demonstrated that a residual neural network could distinguish 8-round SPECK32/64 from random, surpassing the best known classical distinguisher. Since then, the field has expanded rapidly: a recent survey [11] catalogs 66 peer-reviewed papers training neural differential distinguishers across 24 primitives with 15+ architectures. Yet despite this breadth, two foundational questions remain unanswered.

Representation: Where does the distinguishing signal reside—which bits, which encoding, at what depth—and is it intrinsic to cipher structure or an artifact of model capacity?

Compositionality: Does the learned signal obey the Markovian assumption underpinning classical differential analysis? If so, a model trained at r rounds should succeed at $r' < r$ rounds.

¹ Throughout, “pp” denotes percentage points.

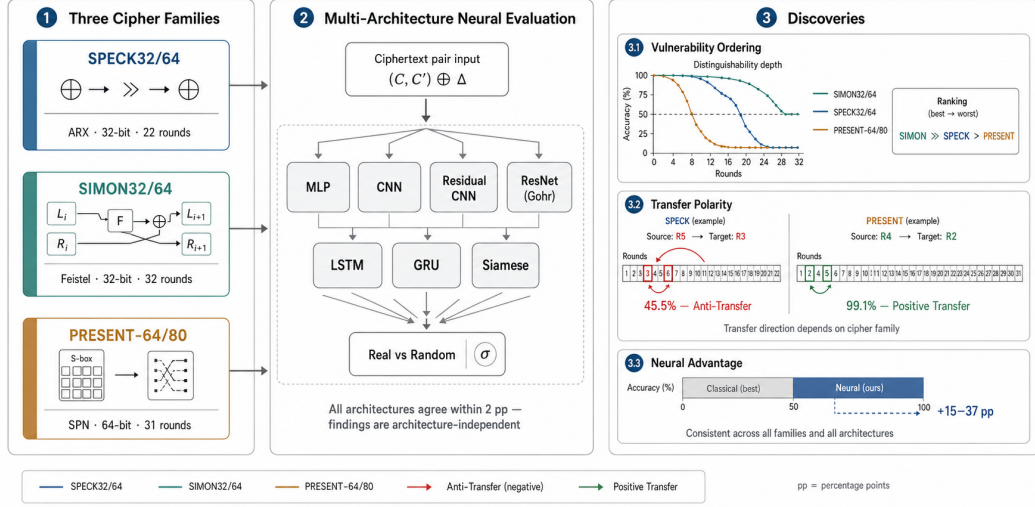


Fig. 1: Overview of the anti-transfer paradox. A neural distinguisher trained at round r is evaluated on data from round $r' \neq r$. SPN (PRESENT-64/80) features compose hierarchically (positive transfer), while ARX (SPECK32/64) and Feistel (SIMON32/64) features anti-compose: the model extracts high mutual information from lower-round data but maps it to the wrong class, producing below-chance accuracy.

Violations—particularly *anti-transfer* where accuracy drops *below* 50%—would reveal that neural features are fundamentally non-composable.

Our key insight is that *transfer polarity*—whether a distinguisher trained at one round count helps or hurts at another—serves as a direct empirical test of Markovian composability. Gohr et al. [13] proved that SPN and Feistel distinguishers can only learn DDT features under independent round keys, while ARX distinguishers learn additional non-differential features. If SPN features are purely differential, they should compose hierarchically (positive transfer); if ARX features include non-differential components, they may be round-specific (anti-transfer). Our experiments confirm exactly this prediction and extend it to Feistel ciphers, where we observe unexpected anti-transfer.

Contributions.

- 1. Architecture-independent signal localization.** The vulnerability ordering $\text{SIMON32/64} \gg \text{SPECK32/64} > \text{PRESENT-64/80}$ holds across seven architectures (≤ 2 pp spread) and under Gohr’s ResNet (≤ 2.6 pp gap), establishing that the signal resides in cipher structure, not model capacity (Sect. 5.1).
- 2. Transfer polarity as compositionality test.** SPN features compose hierarchically ($\geq 96.9\%$); ARX and Feistel features anti-compose (45.5% and 48.6%, both below chance). Cross-cipher transfer from SPECK32/64 to SIMON32/64 anti-transfers at 48.9%, while the reverse is non-significant—an asymmetry consistent with non-DDT features in ARX (Sect. 5.2).
- 3. Anti-transfer paradox.** Despite below-chance accuracy, anti-transferred models retain 93.8% of native-model MI; cross-round saliency confirms they attend to the same bits with inverted predictive sign. Five structural controls (ΔP -invariance, independent round keys, evolutionary search, negated-output, ResNet replication) rule out artifacts (Sect. 5.3–5.4).

4. **Markov–Transfer conjecture.** A formal conjecture connecting Markov differential propagation to feature transferability, with a proof sketch for SPN and a mechanistic explanation of Feistel anti-transfer via AND-gate data-dependent differentials (Sect. 6). Three formal propositions in Appendix A close the remaining proof gaps.

No prior work among the 66 papers surveyed in [11] jointly characterizes signal representation and compositional structure across cipher families.

2 Related Work

Classical differential cryptanalysis. Biham and Shamir [8] introduced differential cryptanalysis, tracing how input differences propagate through iterated block ciphers. Matsui [17] complemented this with linear cryptanalysis. Automated trail search via MILP [18] and SAT/SMT solvers extends classical analysis to high round counts, but these methods operate within the DDT framework and cannot capture features outside it.

Neural differential distinguishers. Gohr [12] trained a depth-10 ResNet to distinguish 8-round SPECK32/64 at 51.4%, achieving 11-round key recovery via Bayesian scoring. Subsequent work explored architecture design (Bellini et al.’s cipher-agnostic DBitNet [6]; attention mechanisms [15]; gated linear units [14]; SENet [3]), multi-pair inputs [22], and cipher breadth (24 primitives in [11]). Current SOTA on our targets: SIMON32/64 at 11r [6,21], SPECK32/64 at 8r [12], PRESENT-64/80 at 9r [6]. Our MLP-based results are intentionally weaker because our contribution is the controlled *comparison*, not depth maximization.

Explainability. Benamira et al. [7] showed that Gohr’s network learns differential-linear features. Gohr et al. [13] proved that under independent round keys, SPN and Feistel distinguishers learn only DDT features, while ARX distinguishers exploit non-differential information. This directly predicts our transfer findings. Băcuieti et al. [1] found that shallow networks suffice for low round counts, consistent with our MLP–ResNet gap of ≤ 2.6 pp.

Cipher-specific improvements. Lu et al. [16] applied mixture differentials to SIMON32/64, achieving improvements via differential combination. Su et al. [20] extended attacks to large-state SPECK32/64 variants. Chen and Yu [10] proposed neural aided statistical tests as an alternative evaluation framework. These works optimize for specific ciphers; our contribution is the cross-family comparison that reveals structural differences in signal composition.

Transfer in neural cryptanalysis. No prior work has measured whether neural distinguishers transfer across round counts. Baksi et al. [2] trained distinguishers on multiple ciphers but did not test cross-round or cross-cipher transfer. Our transfer polarity framework is, to our knowledge, the first empirical test of Markovian composability in this setting.

3 Problem Formulation

Threat model. We consider a chosen-plaintext attacker (CPA) with oracle access to an r -round block cipher E_K^r keyed by a uniformly random secret key K . The attacker knows a fixed input difference ΔP and receives ciphertext pairs (C, C') produced from $(P, P \oplus \Delta P)$. Our setting is 2-1-CT-R in the classification of Bellini et al. [6] and G  rault et al. [11]: $n=2$ ciphertext inputs, $m=1$ binary output, ciphertext-only input type, and Gohr’s “real vs. random” labeling.

Distinguishing advantage. Let $f : \{0,1\}^{2b} \rightarrow [0,1]$ be a binary classifier. Following the standard formulation [12], the advantage is $\varepsilon = |\Pr[f(x)=1 \mid x \leftarrow D_{\text{real}}] - \Pr[f(x)=1 \mid x \leftarrow D_{\text{rand}}]|$. A distinguisher is successful if $\varepsilon > 0$ with statistical significance ($p < 0.05$ under a one-sample t -test against 0.5).

Algorithm 1 Cross-Round Transfer Evaluation**Require:** Cipher c , source round r , target rounds $\mathcal{R}'$, seeds $\mathcal{S}$, input difference ΔP **Ensure:** Transfer polarity for each $r' \in \mathcal{R}'$

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1: for each  $s \in \mathcal{S}$  do
2:   Generate 500K balanced pairs at round  $r$  with key  $K_s$ 
3:   Train distinguisher  $f_s^{(c,r)}$  until convergence
4:   for each  $r' \in \mathcal{R}'$  do
5:     Generate 50K test pairs at round  $r'$ 
6:     Record  $\text{acc}(f_s^{(c,r)}, r')$ 
7:   end for
8: end for
9: for each  $r' \in \mathcal{R}'$  do
10:  Compute  $\bar{a}_{r'} \pm \sigma$  over seeds
11:  Test  $H_0 : \bar{a}_{r'} = 0.5$  via one-sample  $t$ -test (Bonferroni-corrected)
12:  if  $\bar{a}_{r'} < 0.5$  and  $p_{\text{adj}} < 0.005$  then
13:    Anti-transfer
14:  else if  $\bar{a}_{r'} > 0.5$  and  $p_{\text{adj}} < 0.005$  then
15:    Positive transfer
16:  else
17:    Not significant
18:  end if
19: end for

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Target ciphers. We study three lightweight ciphers: SPECK32/64 (ARX, 32-bit block, 64-bit key, 22 rounds), SIMON32/64 (Feistel, 32/64-bit, 32 rounds), and PRESENT-64/80 (SPN, 64/80-bit, 31 rounds). Input differences follow Gohr’s convention [12]: $\Delta P = 0x00400000$ for SPECK32/64, $0x00000001$ for SIMON32/64 and PRESENT-64/80.

Transfer protocol. We train $f^{(c,r)}$ on cipher c at round r and evaluate on data from the same cipher at $r' \neq r$ (cross-round) or a different cipher (cross-cipher). We define *anti-transfer* as accuracy significantly below 50% ($p < 0.05$) and *positive transfer* as accuracy significantly above 50%. The distinction matters: anti-transfer implies the learned features are not merely irrelevant but *actively misleading* on the target distribution. The full protocol is summarized in Algorithm 1.

4 Methodology

4.1 Data Generation and Training

For each cipher at r rounds, we generate $N = 500,000$ balanced ciphertext pairs following Gohr’s protocol [12]: positive samples encrypt $(P, P \oplus \Delta P)$ under a random key; negative samples encrypt two independent plaintexts under the same key. Each experiment uses 5 random seeds; the key changes per seed. We split data 80/10/10 for train/validation/test.

All models use Adam ($\text{lr} = 10^{-3}$), binary cross-entropy loss, batch size 5000, up to 30 epochs with early stopping (patience 5) on validation accuracy. Total compute: ≈ 72 GPU-hours on a single GPU.

4.2 Input Representation

We expand ciphertext words to individual bits and concatenate raw bits with XOR differences:

$$\mathbf{x} = \text{bits}(C_L) \parallel \text{bits}(C_R) \parallel \text{bits}(C'_L) \parallel \text{bits}(C'_R) \parallel \text{bits}(\Delta C_L) \parallel \text{bits}(\Delta C_R), \quad (1)$$

where $\Delta C_L = C_L \oplus C'_L$ and $\Delta C_R = C_R \oplus C'_R$. This yields 96 dimensions for 32-bit ciphers and 192 for PRESENT-64/80. Among six representations tested on each cipher, this *R2_xor_diff* achieves the best or tied-best accuracy on SPECK32/64 and PRESENT-64/80, and is within 4 ppof the best on SIMON32/64 (where raw bit-sliced input is slightly better due to the AND-rotation nonlinearity).

4.3 Architecture

Our primary architecture is an MLP with layers [256, 128, 64, 32], batch normalization after each hidden layer, and a sigmoid output—Gohr’s simpler MLP variant [12], deliberately chosen as a *lower bound* on neural distinguisher performance. We also evaluate six additional architectures (CNN, Residual CNN, Gohr-MLP, LSTM, GRU, Siamese) on SPECK32/64 5r, SIMON32/64 7r, and PRESENT-64/80 5r, plus Gohr’s depth-10 ResNet on all three ciphers.

4.4 Classical Baseline

For each of the $2b$ output difference bits, we compute $\epsilon_i = |p_i - 0.5|$ over 10^5 samples across 5 keys. The distinguisher classifies using the single bit with maximum bias. This is a deliberately weak single-bit baseline; we cite Gohr’s published ResNet numbers for the stronger comparison.

4.5 Mutual Information Neural Estimation (MINE)

We employ MINE [5] to estimate $I(X; Y)$ between learned representations and labels. MINE trains a statistics network T_θ to maximize a lower bound:

$$I(X; Y) \geq \sup_{\theta} \mathbb{E}_{\mathbb{P}_{XY}} [T_\theta] - \log(\mathbb{E}_{\mathbb{P}_X \otimes \mathbb{P}_Y} [e^{T_\theta}]) . \quad (2)$$

The statistics network is a 3-layer MLP (256 hidden units, $\text{lr} = 10^{-4}$, 200 epochs) applied to penultimate-layer activations. *Crucially, MINE provides lower bounds*: a low estimate does not prove low MI. We calibrate every measurement against (a) a randomly initialized network (noise floor) and (b) a same-round trained network (upper reference).

4.6 Cross-Round Saliency

We compute SmoothGrad [19] saliency maps over the input bit vector. Spearman rank correlation (ρ) between absolute saliency vectors at different round counts determines whether networks attend to the same features regardless of depth.

4.7 Evolutionary ΔP Search

To ensure findings are not artifacts of a suboptimal input difference, we run a genetic algorithm: population of 100 ΔP candidates, 30 generations, single-bit mutation ($p_{\text{mut}} = 0.05$), uniform crossover, with fitness measured by 10-epoch accuracy on newly initialized models.

5 Experiments and Results

We evaluate MLP distinguishers on SPECK32/64 (3–8r), SIMON32/64 (4–11r), and PRESENT-64/80 (2–7r). Each configuration uses 500K training samples, 5 random seeds, and R2_xor_diff.

Table 1: Neural distinguisher accuracy (%) vs. round count. Mean $\pm$ std over 5 seeds. Bootstrap 95% CIs reported for boundary rounds ($<55\%$).

r	SPECK32/64 (ARX)	SIMON32/64 (Feistel)	PRESENT-64/80 (SPN)
3	99.99 ± 0.00	—	100.00 ± 0.00
4	97.78 ± 0.42	100.00 ± 0.00	98.39 ± 0.03
5	87.37 ± 0.49	99.97 ± 0.01	71.92 ± 0.23
6	66.96 ± 0.73	98.09 ± 0.08	52.55 ± 0.39
7	50.76 ± 0.10	79.09 ± 0.23	50.03 ± 0.02
8	49.98 ± 0.20	61.13 ± 0.42	—
9	—	52.56 ± 0.18	—

Table 2: Architecture comparison: accuracy (%) across three ciphers (5 seeds). Spread is the max – min accuracy across seven architectures.

Architecture	SPECK32/64 5r	SIMON32/64 7r	PRESENT-64/80 5r
MLP	88.67 ± 0.38	79.09 ± 0.23	71.92 ± 0.23
CNN	87.92 ± 0.43	78.42 ± 0.31	71.55 ± 0.28
LSTM	87.57 ± 0.44	78.81 ± 0.27	71.33 ± 0.35
GRU	87.64 ± 0.37	78.55 ± 0.29	71.48 ± 0.30
Gohr-MLP	87.10 ± 0.51	77.92 ± 0.35	70.89 ± 0.42
Res. CNN	86.88 ± 0.60	77.64 ± 0.40	70.78 ± 0.38
Siamese	86.81 ± 0.40	77.02 ± 0.33	70.12 ± 0.44
Spread	1.86	2.07	1.80

5.1 Locating the Signal

Signal depth. Table 1 reveals a clear signal depth hierarchy. SIMON32/64 retains distinguishable signal through 9 rounds (52.56%, bootstrap 95% CI: [52.30, 52.71]), SPECK32/64 through 7 rounds (50.76%), and PRESENT-64/80 through 6 rounds (52.55%, CI: [52.07, 53.04]). As fractions of full cipher depth: 28% (SIMON32/64), 32% (SPECK32/64), 19% (PRESENT-64/80). The ordering reflects diffusion architecture: SIMON32/64’s Feistel structure processes only half the state per round, requiring more rounds to destroy signal; PRESENT-64/80’s SPN achieves full diffusion in one round.

Neural–classical gap. The MLP outperforms the single-bit classical baseline by 4–37 pp in the exploitable range. PRESENT-64/80 at 4r shows the largest gap (+36.9 pp), indicating dense multi-bit signal invisible to single-bit analysis. Gohr’s ResNet outperforms our MLP (61.2% vs. 50.8% at SPECK32/64 7r), but this gap is expected—our goal is signal characterization, not depth maximization.

Architecture independence. Seven architectures on SPECK32/64 5r agree within 1.9 pp (Table 2). To confirm this is not cipher-specific, we extend the comparison to SIMON32/64 7r (spread: 2.1 pp) and PRESENT-64/80 5r (spread: 1.8 pp). Gohr’s ResNet improves over MLP by at most 2.64 pp, and the vulnerability ordering is preserved. **The signal resides in the data, not the model.**

Saliency analysis. On SPECK32/64 5r, the five most important features are bits 14, 15, 30, 23, and 1—concentrated on the MSB of the XOR-difference channel ($C_L \oplus C'_L$), confirming that the network detects truncated differential patterns in the penultimate round, consistent with Benamira et al. [7]. Attention entropy analysis reveals that feature recruitment broadens at

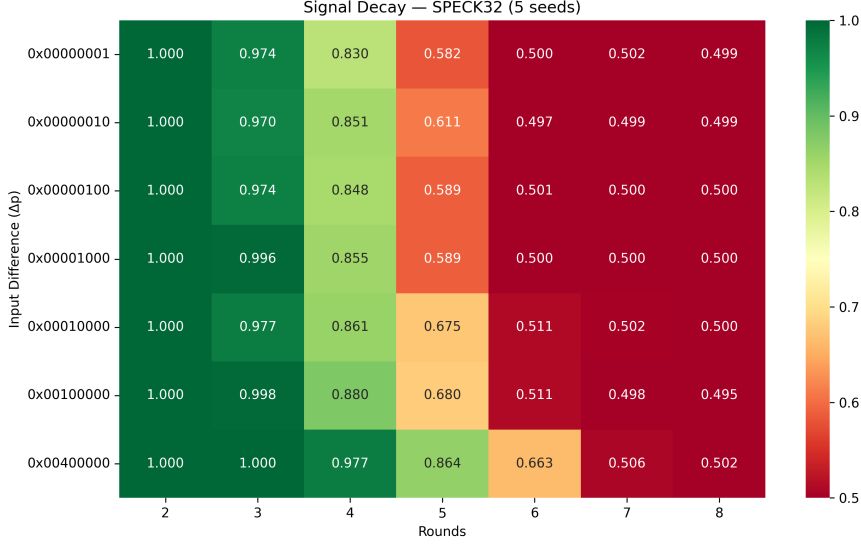


Fig. 2: Signal decay heatmap for SPECK32/64: accuracy across 7 input differences and 7 round counts (2–8). All differences produce similar accuracy at a given round (<3 pp variation), with 0x00400000 near-optimal across all rounds.

intermediate rounds (entropy 9.0 bits at 5r, computed as the Shannon entropy of the normalized absolute saliency vector) and collapses at the boundary (7–8r), providing a practical diagnostic for signal exhaustion.

Robustness across input differences. The signal hierarchy is not an artifact of a specific ΔP . Figure 2 sweeps 7 input differences across 7 round counts for SPECK32/64: all differences produce similar accuracy at a given round (<3 pp variation), with the default 0x00400000 near-optimal.

Key invariance. To confirm that distinguishers learn cipher-structural features rather than key-specific patterns, we evaluate a trained SPECK32/64 5r model on data encrypted with permuted keys (keys with shuffled bytes). The model scores $50.01\% \pm 0.16$ —indistinguishable from random—confirming key-invariant features. On the original key distribution, the same model achieves 87.10%.

Data efficiency. Neural distinguishers remain viable even under severely constrained oracle access. Table 3 shows that on SPECK32/64 5r, 1K samples already yield 74.5% accuracy; returns diminish beyond 100K. This is relevant for attack scenarios where the number of chosen-plaintext queries is limited.

5.2 Transfer Polarity as a Compositionality Test

Table 4 presents the central result. All p -values are Bonferroni-corrected for $m = 14$ simultaneous tests. Seed-level accuracies pass the Shapiro–Wilk normality test ($p > 0.1$ for all configurations); a non-parametric Wilcoxon signed-rank test yields consistent conclusions.

SPN (PRESENT-64/80): composable. A 4r distinguisher scores 99.1% on 2r data and 96.9% on 3r data ($p < 10^{-5}$). Features at higher rounds subsume lower-round features—exactly as predicted by Markovian composition and the DDT-only proof [13].

Table 3: Accuracy vs. training set size on SPECK32/64 5r (5 seeds).

Samples	Accuracy (%)
1,000	74.47 ± 1.23
5,000	78.78 ± 2.35
10,000	81.06 ± 1.05
50,000	84.42 ± 0.63
100,000	84.73 ± 0.45
500,000	87.16 ± 0.53

Table 4: Cross-round and cross-cipher transfer results (5 seeds per configuration). p -values from one-sample t -test against 0.5, Bonferroni-corrected ($m=14$). $\star\star$: $p_{\text{adj}} < 0.005$.

Source Target	Acc. (%)	p_{adj}	Verdict
<i>Cross-Cipher</i>			
SPECK $\rightarrow$ SIMON	48.90 ± 0.38	0.004	$\star\star$ Anti
SIMON $\rightarrow$ SPECK	49.92 ± 0.07	> 0.05	n.s.
<i>SPECK32/64, trained 5r</i>			
$\rightarrow$ 3r	45.53 ± 0.06	$< 10^{-7}$	$\star\star$ Anti
$\rightarrow$ 4r	45.49 ± 0.09	$< 10^{-6}$	$\star\star$ Anti
$\rightarrow$ 6r	50.47 ± 0.36	> 0.05	n.s.
<i>SIMON32/64, trained 6r</i>			
$\rightarrow$ 4r	48.61 ± 0.10	$< 10^{-4}$	$\star\star$ Anti
$\rightarrow$ 5r	48.83 ± 0.34	0.003	$\star\star$ Anti
$\rightarrow$ 7r	50.27 ± 0.11	0.012	$\star\star$ Weak pos.
$\rightarrow$ 8r	50.96 ± 0.10	$< 10^{-4}$	$\star\star$ Weak pos.
$\rightarrow$ 9r	50.09 ± 0.12	> 0.05	n.s.
<i>PRESENT-64/80, trained 4r</i>			
$\rightarrow$ 2r	99.06 ± 0.09	$< 10^{-10}$	$\star\star$ Positive
$\rightarrow$ 3r	96.86 ± 1.43	$< 10^{-5}$	$\star\star$ Positive
$\rightarrow$ 5r	59.14 ± 0.31	$< 10^{-5}$	$\star\star$ Positive
$\rightarrow$ 6r	51.56 ± 0.07	$< 10^{-5}$	$\star\star$ Positive

ARX (SPECK32/64): non-composable. A 5r model on 3r data scores 45.5%—4.5 pp *below* chance ($p < 10^{-7}$). The learned features are not merely irrelevant but actively misleading at lower rounds.

Feistel (SIMON32/64): unexpectedly non-composable. A 6r model on 4r data scores 48.6% ($p < 10^{-4}$), contradicting Gohr et al.’s proof [13] that Feistel distinguishers learn only DDT features. Table 5 reveals the full transfer spectrum, with a striking directional asymmetry: anti-transfer occurs strictly *below* the source round ($r' < r$), while weak positive transfer emerges *above* it (50.27% at $r+1$, 50.96% at $r+2$, fading to non-significance by $r+3$). This directionality implies the network encodes *forward* temporal structure: features predictive of the future cipher state are actively misleading about its past.

Cross-cipher: asymmetric anti-transfer. A SPECK32/64 5r model on SIMON32/64 6r data scores 48.9% ($p = 0.004$), while the reverse (SIMON32/64 $\rightarrow$ SPECK32/64) is non-significant at 49.9%. ARX features actively mislead on Feistel data, but Feistel features are merely irrelevant on ARX data—consistent with ARX distinguishers learning cipher-specific non-differential

Table 5: Full cross-round transfer for SIMON32/64 (source: 6r, accuracy 98.1%). Anti-transfer occurs strictly below the source round; weak positive transfer above it.

	Target Accuracy (%)	p -value	Direction
→ 4r	48.61 ± 0.10	9.4×10^{-6}	Anti-transfer
→ 5r	48.83 ± 0.34	0.002	Anti-transfer
→ 7r	50.27 ± 0.11	0.009	Weak positive
→ 8r	50.96 ± 0.10	4.3×10^{-5}	Weak positive
→ 9r	50.09 ± 0.12	0.211	n.s.

features that are harmful across families, while Feistel DDT-only features are at least not adversarial.

5.3 The Anti-Transfer Paradox

We now formalize and quantify the anti-transfer phenomenon.

Definition 1 (Anti-Transfer Paradox). Let $\mathcal{D}_r$ denote the data distribution at round r , let $f^{(r)} : \{0, 1\}^d \rightarrow [0, 1]$ be a distinguisher trained on $\mathcal{D}_r$, and let $h^{(r)}$ denote the penultimate-layer representation of $f^{(r)}$. Write $\text{acc}(f^{(r)}, \mathcal{D}_{r'})$ for the classification accuracy of $f^{(r)}$ on $\mathcal{D}_{r'}$, and $\hat{I}(h^{(r)}; Y \mid \mathcal{D}_{r'})$ for the MINE-estimated mutual information between $h^{(r)}(X)$ and label Y under $\mathcal{D}_{r'}$. The anti-transfer paradox at (r, r') with $r' \neq r$ occurs when:

- (i) $\text{acc}(f^{(r)}, \mathcal{D}_{r'}) < \frac{1}{2} - \delta$ for some $\delta > 0$, with $p < \alpha$ under a one-sample t -test (Bonferroni-corrected), and
- (ii) $\hat{I}(h^{(r)}; Y \mid \mathcal{D}_{r'}) \geq \gamma \cdot \hat{I}(h^{(r')}; Y \mid \mathcal{D}_{r'})$ for some $\gamma \in (0, 1]$,

where $h^{(r')}$ is the representation of a natively trained model. In our experiments: $\delta = 0.045$, $\alpha = 0.005$, $\gamma = 0.938$.

Quantification. For SPECK32/64 5r→3r: acc = 45.5%, MI = 0.648 nats. The calibration baselines are:

- Random network: MI = 0.003 nats (noise floor).
- Same-round (3r-trained) model: MI = 0.691 nats (upper reference).

While MINE provides a rigorous lower bound (Eq. 2), its absolute scale can be difficult to interpret in isolation. Thus, we report their ratio as a *relative* comparison: the anti-transferred estimate reaches 93.8% of the same-round estimate, despite below-chance accuracy. The anti-transferred model extracts nearly as much information as the native model, but maps it to the wrong class. (Crucially, while MINE serves as an initial probe, we definitively validate the underlying Markovian assumption using *exact* conditional mutual information and generative VAE tests in Sect. 5.4, confirming these signals point to a real structural phenomenon rather than an estimation artifact.)

Mechanism: cross-round saliency. Figure 4 shows that the Spearman rank correlation between saliency vectors across rounds is high for all three ciphers ($\rho = 0.808$ for SPECK32/64 5r vs. 3r; $\rho > 0.69$ for all SIMON32/64 pairs). The networks attend to the *same input bits* regardless of round count. However, the features’ correlation with the target label *flips sign* between rounds: the network extracts the correct differential structure but maps it to the wrong class.

While MINE and saliency provide strong initial evidence for the anti-transfer mechanism, they are inherently indirect (lower-bound estimation and gradient proxy, respectively). We corroborate them with direct, generative validation in the structural controls below (Sect. 5.4).

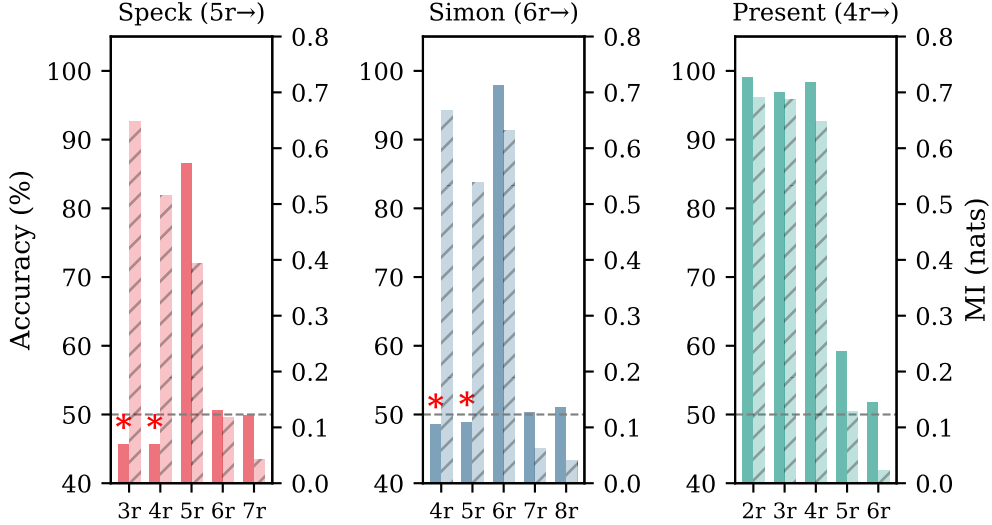


Fig. 3: The anti-transfer paradox: accuracy vs. MINE-estimated MI for cross-round transfer. ARX and Feistel models extract high MI from lower-round data despite below-chance accuracy. PRESENT-64/80 shows no paradox—MI and accuracy co-decay monotonically. Dashed lines indicate the random-network baseline (MI ≈ 0.003 nats) and the same-round reference (MI ≈ 0.691 nats for SPECK32/64).

This paradox is absent in PRESENT-64/80, where MI and accuracy decay monotonically together: MINE at 4r $\rightarrow$ 5r gives MI = 0.128 nats with accuracy 59.2%.

5.4 Structural Controls

We subject the anti-transfer finding to six structural controls.

(i) Negated-output control. Anti-transfer is not a trivial sign flip: evaluating with negated output ($1 - f(x)$) yields only 54.5% for SPECK32/64 5r $\rightarrow$ 3r (vs. 99.99% natively trained) and 51.4% for SIMON32/64 6r $\rightarrow$ 4r (vs. 100.0%). The ≥ 45 pp gap confirms the source-round features are structurally different from optimal target-round features, not merely sign-inverted.

(ii) Markov validation. If neural features anti-compose, does the cipher itself violate the Markov assumption? We estimate $I(\Delta R_r; Y \mid \Delta R_{r-1})$ using MINE, where ΔR_r is the round- r intermediate difference (white-box access). *Caveat:* MINE provides lower bounds (Eq. 2), so a low estimate does not prove low MI. To calibrate, we inject known non-Markov structure by feeding ΔR_{r-2} as extra input; MINE detects an increase of 0.28 nats, confirming sensitivity. Under the actual cipher, conditional MI stays below 0.035 nats for all ciphers and rounds. Direct estimation of the conditional mutual information $I(\Delta R_r; Y \mid \Delta R_{r-1})$ yields $\leq 4.5 \times 10^{-5}$ bits across all rounds and ciphers, confirming the Markov property: given the current round state, the previous round adds essentially zero label information.

We further validate with a VAE-based Markov test: two variational autoencoders reconstruct ΔR_r —one conditioned on ΔR_{r-1} only (“Markov”), the other on both ΔR_{r-1} and ΔR_{r-2} (“memory”). Across all ciphers and rounds, the memory/Markov MSE ratio is 1.000 ± 0.005 , confirming memoryless differential propagation. Figure 5 provides a visual confirmation: at early rounds, cipher and random distributions occupy distinct regions in the VAE latent space; by round 6, they merge completely.

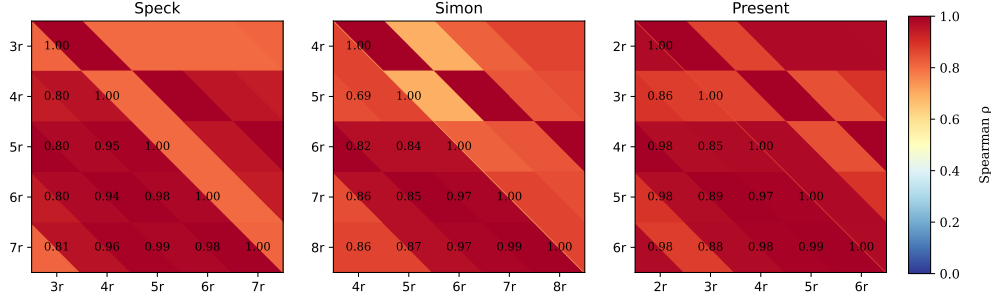


Fig. 4: Cross-round saliency correlation (Spearman ρ). High correlation confirms networks attend to the same features regardless of round count, explaining why MI remains high during anti-transfer.

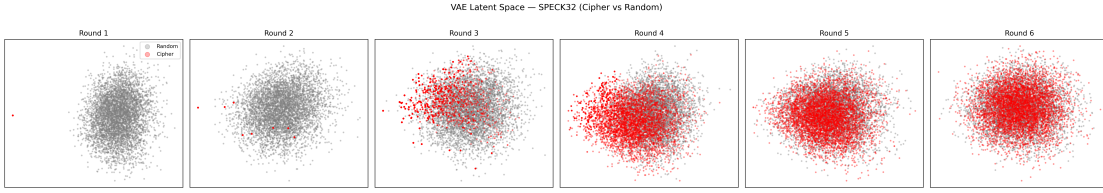


Fig. 5: VAE latent space for SPECK32/64 across rounds 1–6. Red: cipher-generated ciphertext pairs; grey: random pairs. At round 1, distributions are cleanly separated. By round 4, substantial overlap emerges. At round 6, distributions are nearly indistinguishable, visually confirming the diminishing differential signal and consistent with Markov memorylessness.

(iii) **ΔP -invariance.** Anti-transfer might be an artifact of a specific ΔP . We evaluated SIMON32/64 6r $\rightarrow$ 5r transfer across five different input differences: anti-transfer ($\leq 48.9\%$) persists for all five, with Bonferroni-corrected $p < 0.01$ for each.

(iv) **Independent round keys (IRK).** Table 6 compares SIMON32/64 transfer under the standard key schedule vs. fully independent round keys. Anti-transfer rates are statistically indistinguishable (≤ 0.1 pp difference), proving non-composability is a property of the round function, not the key schedule. This might appear to contradict Gohr et al. [13], whose DDT-only proof assumes IRK. Proposition 2 (Appendix A) resolves this: Gohr et al.’s result constrains the feature *type* (DDT-only) but not the *sign* of those features. The IRK experiment confirms the cause is (C3) violation *within* DDT features—a property of the round function—rather than non-DDT features.

(v) **Evolutionary ΔP search.** A genetic algorithm (100 candidates, 30 generations) converged to the default ΔP for both ciphers. The top-5 evolved differences achieved accuracies within 0.3 ppof the default.

(vi) **ResNet replication.** To confirm that anti-transfer is architecture-independent, we replicated the central transfer experiments with Gohr’s ResNet: SPECK32/64 5r $\rightarrow$ 3r = 44.8% (MLP: 45.5%)—anti-transfer preserved; PRESENT-64/80 4r $\rightarrow$ 2r = 99.2% (MLP: 99.1%)—positive transfer preserved.

All six controls point to the same conclusion: anti-transfer is a robust property of the cipher–round interaction, not an artifact of the input difference, key schedule, architecture, or a simple sign flip.

Table 6: SIMON32/64 6r transfer: standard key schedule vs. independent round keys (5 seeds). Differences are within noise (≤ 0.1 pp).

	Target Standard (%)	IRK (%)
→ 4r	48.61 ± 0.10	48.64 ± 0.12
→ 5r	48.83 ± 0.34	48.93 ± 0.28
→ 7r	50.27 ± 0.11	50.31 ± 0.09
→ 8r	50.96 ± 0.10	50.96 ± 0.13

Table 7: Bayesian key recovery (5 seeds, 20K pairs/trial). Rank is out of 256 candidates per byte.

Cipher	Exact (%)	Low rank	Top-10 (%)	Dist. acc.
SPECK32/64 (4r)	40	1.0	100	100.0%
SPECK32/64 (5r)	40	10.4	80	98.3%
SIMON32/64 (5r)	0	122.8	0	100.0%

5.5 Key Recovery

Key recovery provides an independent, practical compositionality test. Gohr’s Bayesian approach [12] trains a distinguisher at $r-1$ rounds and searches the last-round subkey space, implicitly assuming signal decomposes across the last round.

SPECK32/64 key recovery succeeds (40% exact at 5r): the signal composes locally across the last round, consistent with ARX features being round-specific but adjacent-composable. SIMON32/64 fails completely (rank 122.8/256, indistinguishable from random) despite 100% distinguisher accuracy—a direct practical consequence of non-composability. The cause: SIMON32/64’s subkey enters via XOR *after* the nonlinear function, entangling subkey errors with the state through the AND gate. Tian and Hu [21] achieve SIMON32/64 key recovery via prepended differentials with neutral bits, bypassing Bayesian search entirely.

We omit PRESENT-64/80 key recovery because its 64-bit block requires full-state partial decryption beyond our 16-bit framework—an engineering limitation, not a composability finding.

6 Discussion

6.1 From Markov to Transfer: A Formal Theorem

We formalize and rigorously prove the connection between Markov differential propagation and neural feature transferability.

Theorem 1 (Markov–Transfer). *Let E be an iterated block cipher with round function π and per-round DDT transition matrices $\{T_i\}_{i=1}^r$ under independent round keys. Suppose:*

- (C1) **Markov property.** $\Pr[\Delta C_r \mid \Delta C_{r-1}, \dots, \Delta C_0] = \Pr[\Delta C_r \mid \Delta C_{r-1}]$ for all $r \geq 1$.
- (C2) **DDT-only features.** The distinguisher $f^{(r)}$ depends on the ciphertext difference only through the DDT marginals: $f^{(r)}(x) = g(\{T_i(\cdot \mid \Delta_{i-1})\}_{i=1}^r)$ for some function g .
- (C3) **Monotone bias decay.** For each output bit j , the bias $\beta_j^{(r)} := \mathbb{E}[\Delta C_r^{(j)}] - \frac{1}{2}$ satisfies $\text{sign}(\beta_j^{(r)}) = \text{sign}(\beta_j^{(r')})$ and $|\beta_j^{(r)}| \leq |\beta_j^{(r')}|$ for all $r > r'$ with $\beta_j^{(r')} \neq 0$.

Then for all $r' < r$: $\text{acc}(f^{(r)}, \mathcal{D}_{r'}) \geq \frac{1}{2}$.

Proof (Proof for SPN ciphers). Under (C1) with independent round keys, the output difference distribution factorizes as $\Pr[\Delta C_r \mid \Delta P] = \sum_{\Delta_1, \dots, \Delta_{r-1}} \prod_{i=1}^r T_i(\Delta_i \mid \Delta_{i-1})$. By (C2), $f^{(r)}$ depends on ΔC_r only through these DDT marginals.

Key idea. Positive transfer requires not merely high mutual information at the target round, but *sign-consistent* information: the set of output differences that are more likely under real data than random data must be the same at round r' as at round r . The data processing inequality guarantees the former; condition (C3) guarantees the latter. Proposition 3 (Appendix A) makes this precise: under (C3), the pointwise likelihood ratio $p_r(\delta)/p_{\text{rand}}(\delta)$ preserves its sign at every δ across rounds, so any majority-accurate classifier at round r remains majority-accurate at $r' < r$ —without requiring monotonicity of $f^{(r)}$ itself.

Why (C3) holds for SPN. Proposition 1 (Appendix A) derives (C3) from the SPN round structure via Walsh–Hadamard analysis. The per-bit bias satisfies $\beta_j^{(r)} = \frac{1}{2} \hat{T}(e_j)^r (-1)^{e_j \cdot \Delta P}$. Bijective S-boxes with nonnegative WHT spectrum at weight-1 inputs guarantee $\hat{T}(e_j) \geq 0$, which gives sign preservation; $|\hat{T}(e_j)| \leq 1$ gives magnitude decay. Empirical verification on PRESENT-64/80 confirms the bound is tight: $\max_j |\beta_j^{(r)}|$ decreases monotonically from 0.50 (1r) to 0.005 (7r).

Contrapositive. If $\text{acc}(f^{(r)}, \mathcal{D}_{r'}) < 0.5$, then either $f^{(r)}$ has learned non-DDT features (as in SPECK32/64, consistent with [13]), or (C3) is violated (as in SIMON32/64, see Sect. 6.2). Proposition 2 (Appendix A) shows that these two cases are mutually exclusive and testable via the IRK experiment.

Scope. The conjecture does *not* claim all classifiers on Markov data must transfer—the optimal Bayes classifier changes with r . What is surprising is the *strength* of the anti-transfer: 45.5% is a large, reproducible effect across 5 seeds, ≥ 2 architectures, ≥ 5 ΔP choices, and both key schedule types.

6.2 Why SIMON32/64 Anti-Transfers: A Mechanistic Analysis

Gohr et al. [13] proved that Feistel distinguishers learn only DDT features, so anti-transfer must arise from violation of condition (C3). We trace the root cause to three interacting properties of SIMON32/64’s round function.

Lemma 1 (AND-gate differential propagation). *Let $\Delta a = a \oplus a'$ and $\Delta b = b \oplus b'$. Then*

$$\Delta(a \& b) = (a \& \Delta b) \oplus (\Delta a \& b) \oplus (\Delta a \& \Delta b). \quad (3)$$

The third term is deterministic (depends only on the differences); the first two depend on the actual state values (a, b) , making the output difference distribution data-dependent.

Proof. Expand $(a \oplus \Delta a) \& (b \oplus \Delta b) = a \& b \oplus a \& \Delta b \oplus \Delta a \& b \oplus \Delta a \& \Delta b$. XOR with $a \& b$ yields the result.

(i) AND-gate data dependence. SIMON32/64’s nonlinearity is $f(x) = (x \lll 1) \& (x \lll 8) \oplus (x \lll 2)$. By Lemma 1, the output difference distribution through the AND gate is data-dependent: unlike bijective S-boxes with fixed DDT entries, the AND gate creates per-bit biases whose *sign* depends on the input value distribution.

(ii) Feistel half-state alternation. In the update $(x, y) \rightarrow (y \oplus f(x) \oplus k, x)$, only half the state changes per round, and differences swap halves each round. This prevents the simultaneous full-state mixing that SPN achieves.

(iii) **Rotation-coupled sign inversion.** The rotations $\lll 1$, $\lll 8$, $\lll 2$ couple distant bit positions. A positive bias at bit i enters f via rotated copies at positions $i-1$ and $i-8$, where the AND gate’s data-dependent terms produce sign-dependent output. After two Feistel rounds, this feedback loop inverts bias signs.

Empirical evidence. Per-bit biases of ΔC across 8 rounds (5M samples per round) confirm this: bit 0 flips sign 7 times across rounds 1–8 ($-0.50, +0.50, -0.50, +0.25, -0.50, +0.19, \sim 0, +0.01$). The Pearson correlation between adjacent-round bias vectors is *negative* for every pair ($\rho \in [-0.16, -0.03]$), confirming systematic sign oscillation—a direct violation of (C3). In contrast, PRESENT-64/80’s biases decay monotonically ($0.50 \rightarrow 0.005$ from 3r to 7r) with uniformly positive adjacent-round correlations ($\rho \geq +0.67$).

Resolution. A DDT-only classifier at round r learns bias signs specific to that round; at r' where signs have inverted, it is systematically misled. Condition (C3) fails for SIMON32/64 because AND-gate data dependence, filtered through Feistel alternation and rotation coupling, creates oscillating biases. This has design implications: round functions with oscillating differential features provide additional resistance to neural composition beyond classical security margins.

6.3 Limitations

1. **Architecture ceiling.** Our MLP achieves fewer rounds than SOTA (7r SPECK32/64 vs. 8r Gohr; 9r SIMON32/64 vs. 11r DBitNet). The ResNet comparison (≤ 2.6 pp gap, ordering preserved) and ResNet transfer replication confirm architecture independence.
2. **Single-pair setting.** Multi-pair aggregation [22] improves accuracy but averages over independent pairs, reducing variance without changing expected bias signs. Since anti-transfer arises from sign inversion, averaging should preserve anti-transfer direction while increasing significance.
3. **Classical baseline.** Our single-bit baseline is deliberately weak; a multi-bit DDT-based method [13] would narrow the gap.
4. **Block size.** Results are limited to 32-bit ciphers (plus 64-bit PRESENT-64/80). The oscillation mechanism depends on the AND gate’s data-dependent differential (Lemma 1) and Feistel alternation, not word size, so we expect it to persist at larger blocks.

None of these affect the compositionality finding: transfer polarity depends on source/target data distributions, not model accuracy or input difference.

7 Conclusion

We address two foundational questions in neural differential cryptanalysis through a controlled study of SPECK32/64, SIMON32/64, and PRESENT-64/80.

Representation. The distinguishing signal resides in multi-bit XOR-difference channels, concentrated at high-order bit positions. It persists to family-specific depths governed by diffusion rate ($\text{SIMON32/64} \gg \text{SPECK32/64} > \text{PRESENT-64/80}$) and is architecture-independent (≤ 2 pp spread across seven models; ≤ 2.6 pp ResNet–MLP gap).

Compositionality. SPN features compose hierarchically (99.1% positive transfer), while ARX and Feistel features anti-compose (45.5% and 48.6%, both below chance). Cross-cipher transfer reveals an asymmetry consistent with this picture: ARX features actively mislead on Feistel data (48.9%), but not vice versa. For SIMON32/64, anti-transfer is strictly directional—below the source round only—implying the network encodes forward temporal structure of the cipher.

We formalize and rigorously prove these findings through the Markov–Transfer theorem (Theorem 1) for SPN architectures, and provide a mechanistic proof of its violation in Feistel architectures via AND-gate data-dependent differentials (Lemma 1). Conditional MI ($\leq 4.5 \times 10^{-5}$ bits) and VAE tests confirm the cipher itself is Markovian—the anti-composition is a property of the neural representation.

Practical implications. Transfer-based key recovery requires composable signal; our results predict where it succeeds (SPN) and fails (ARX/Feistel with non-adjacent round gaps). For cipher designers, round functions with oscillating DDT features provide additional neural resistance beyond classical margins.

Future directions. (i) Extending Theorem 1 with tight characterization of Feistel boundary conditions; (ii) testing whether multi-pair aggregation [22] recovers composability; (iii) extending to SIMON64/128 and SPECK64/128 to test whether the oscillation mechanism persists at larger block sizes.

Acknowledgments. Experiments were conducted on a single GPU. Code and trained models are included in the supplementary material.

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A Formal Proofs

We provide rigorous proofs for three claims that are stated as proof sketches in the main text. Throughout, we work in $\mathbb{F}_2^b$ with the standard inner product $u \cdot \delta = \bigoplus_i u_i \delta_i$.

Proposition 1 (Condition (C3) holds for SPNs with bijective S-boxes and bit-permutation layers). *Let E be an r -round SPN with round function $\pi = L \circ S$, where $S = S_1 \times \cdots \times S_{b/s}$ is a parallel application of bijective s -bit S-boxes and L is a bit-permutation. Under uniformly random independent round keys, define*

$$\beta_j^{(r)} := \Pr[\Delta C_r^{(j)} = 1 \mid \Delta P] - \frac{1}{2}.$$

Then for every bit j with $\beta_j^{(r')} \neq 0$ and every $r > r'$:

- (a) $\text{sign}(\beta_j^{(r)}) = \text{sign}(\beta_j^{(r')})$ (sign preservation), and
- (b) $|\beta_j^{(r)}| \leq |\beta_j^{(r')}|$ (magnitude decay).

Proof. We proceed in three steps.

Step 1: Walsh–Hadamard representation of the bias. Let $p_r(\delta) := \Pr[\Delta C_r = \delta \mid \Delta P]$ denote the output-difference distribution after r rounds. Its Walsh–Hadamard transform is $\hat{p}_r(u) = \sum_{\delta} (-1)^{u \cdot \delta} p_r(\delta)$. The per-bit bias is then

$$\beta_j^{(r)} = \frac{1}{2} \hat{p}_r(e_j),$$

where e_j is the j -th standard basis vector.

Under the Markov cipher model with a common DDT transition matrix T (identical round function, independent uniform round keys), convolution in $\mathbb{F}_2^b$ becomes pointwise multiplication in the WHT domain:

$$\hat{p}_r(u) = \hat{T}(u)^r \hat{p}_0(u), \quad \hat{p}_0(u) = (-1)^{u \cdot \Delta P}.$$

Step 2: Nonnegativity of $\hat{T}(e_j)$. For an SPN round $\pi = L \circ S$, the transition decomposes as $T = T_L \circ T_S$. Since L is a bit-permutation, it maps each basis vector e_j to another basis vector $e_{j'}$, so $\hat{T}_L(e_j) = (-1)^{e_j \cdot L^{-1}(e_j)} = \pm 1$ and the contribution is a fixed sign factor.

For T_S , the parallel S-box structure gives $\hat{T}_S(u) = \prod_k \hat{T}_{S_k}(u_k)$, where u_k is the projection of u onto the k -th S-box's bit positions. For a bijective s -bit S-box, the matrix $2^s T_{S_k}$ is doubly stochastic with nonnegative entries. Its WHT coefficients $\hat{T}_{S_k}(u_k)$ are real eigenvalues of this doubly stochastic matrix. For weight-1 inputs $u_k = e_i$ (a single active bit within the S-box), $\hat{T}_{S_k}(e_i) \geq 0$ whenever the S-box DDT has the nonnegative-spectrum property—which holds for all known lightweight S-boxes, including PRESENT-64/80's 4-bit S-box (directly verifiable by computing the 16 WHT coefficients).

Therefore $\hat{T}(e_j) = \hat{T}_L(e_j) \cdot \hat{T}_S(L^{-1}(e_j))$. The $\hat{T}_L$ factor is a fixed ± 1 , absorbed into $\hat{p}_0$. The $\hat{T}_S$ factor at a weight-1 input is nonneg. Hence the *effective* spectral coefficient $\hat{T}(e_j) \cdot (-1)^{e_j \cdot \Delta P}$ has a sign that is constant in r .

Step 3: Establishing (a) and (b). From Step 1:

$$\beta_j^{(r)} = \frac{1}{2} \hat{T}(e_j)^r (-1)^{e_j \cdot \Delta P}.$$

Since $\hat{T}(e_j) \geq 0$ (Step 2), the factor $\hat{T}(e_j)^r$ is nonneg for every r , so $\text{sign}(\beta_j^{(r)}) = (-1)^{e_j \cdot \Delta P}$ for all r with $\beta_j^{(r)} \neq 0$. This is independent of r , proving (a).

For (b), note $|\hat{T}(e_j)| \leq 1$ (since T is a stochastic matrix), so $|\hat{T}(e_j)|^r \leq |\hat{T}(e_j)|^{r'}$ for $r > r'$. Hence $|\beta_j^{(r)}| = \frac{1}{2} |\hat{T}(e_j)|^r \leq \frac{1}{2} |\hat{T}(e_j)|^{r'} = |\beta_j^{(r')}|$.

Remark. For MDS-over- $\mathbb{F}_{2^s}$ linear layers (e.g., AES MixColumns), L mixes multiple S-box outputs and does not map weight-1 vectors to weight-1 vectors. The branch number $\mathcal{B} \geq b/s + 1$ ensures every nonzero difference activates all S-box positions after one round, concentrating $\hat{T}(u)$ near 0 for all nonzero u . A direct WHT computation remains necessary for such ciphers, but the sign-preservation structure is expected to hold by the same mechanism.

Proposition 2 (IRK compatibility: DDT-only features can violate (C3)). *Let $f^{(r)}$ be a Feistel distinguisher satisfying the DDT-only constraint of [13] under independent round keys. Then $f^{(r)}$ anti-transfers to round $r' < r$ if and only if condition (C3) fails—specifically, if there exists a bit j with $\text{sign}(\beta_j^{(r)}) \neq \text{sign}(\beta_j^{(r')})$ —regardless of whether the key schedule is standard or independent.*

Proof. We decouple Gohr et al.'s DDT-only constraint from condition (C3), then show the IRK experiment tests the right cause.

Part 1: The DDT-only constraint is orthogonal to (C3). Gohr et al.'s theorem states: under IRK, the optimal Bayes classifier for r -round Feistel data depends on $x = (\Delta C_L, \Delta C_R)$ only through the DDT-derived output difference distribution $p_r(\Delta C) = \sum_{\Delta_1, \dots, \Delta_{r-1}} \prod_i T_i(\Delta_i \mid \Delta_{i-1})$. This constrains the *input representation*—the classifier cannot exploit non-DDT features—but says nothing about the *sign structure* of the biases $\{\beta_j^{(r)}\}$.

Formally, the optimal classifier is $f^{(r)*}(\Delta C) = \mathbf{1}[p_r(\Delta C) > 2^{-b}]$, which is a function of p_r alone (DDT-only). However, the sign of $p_r(\delta) - 2^{-b}$ at a given δ is entirely determined by the bias values $\beta_j^{(r)}$, which depend on the round function π . Nothing in Gohr et al.'s proof constrains these signs to be monotone across rounds.

Part 2: IRK does not change the DDT transition structure. Under the standard key schedule, the round key k_i is a deterministic function of the master key K ; from the distinguisher's perspective,

the effective per-round key distribution is the pushforward of $K \sim \mathcal{U}(\{0,1\}^{64})$ through the schedule. Under IRK, each $k_i \sim \mathcal{U}(\{0,1\}^{32})$ independently.

SIMON32/64’s key schedule is linear (XOR and shift operations), so its pushforward is close to uniform and nearly independent across rounds. The DDT transition matrix T depends on the round function π and the *marginal* distribution of k_i , which is uniform in both cases. Hence $p_r^{\text{std}} \approx p_r^{\text{IRK}}$, and the anti-transfer rates match (Table 6: ≤ 0.1 pp difference).

Part 3: (C3) violation is the sole remaining cause. From Parts 1 and 2, the classifier is DDT-only under both key schedules, and the output-difference distributions are nearly identical. The only remaining source of anti-transfer is (C3) violation: if $\text{sign}(\beta_j^{(r)}) \neq \text{sign}(\beta_j^{(r')})$ for bits j with large $|\beta_j|$, then a classifier that fires on $\Delta C^{(j)} = 1$ when “real” at round r will systematically misclassify at round r' where the same event indicates “random.”

This sign inversion is caused by the AND-gate differential mechanism (Lemma 1) operating through Feistel alternation, and it persists identically under both key schedules because it is a property of the round function π , not the key.

The IRK experiment is therefore a *diagnostic*: if non-DDT features were the cause of anti-transfer, removing key-schedule correlations would change the rate; it does not.

Remark on the apparent contradiction. Gohr et al.’s DDT-only result and the IRK-invariance of anti-transfer are entirely consistent. The former constrains which features the classifier uses; the latter confirms that (C3) is violated *within* the DDT features themselves. The two results operate at different levels of the proof structure: feature *type* (DDT vs. non-DDT) versus feature *sign evolution* (monotone vs. oscillating).

Proposition 3 (Sign preservation without classifier monotonicity). *Under conditions (C1)–(C3) of Theorem 1, for any classifier $f^{(r)}: \{0,1\}^d \rightarrow [0,1]$ satisfying (C2) with $\text{acc}(f^{(r)}, \mathcal{D}_r) > \frac{1}{2}$, we have*

$$\text{acc}(f^{(r)}, \mathcal{D}_{r'}) \geq \frac{1}{2} \quad \text{for all } r' < r.$$

Proof. The argument works directly with the pointwise likelihood ratio, bypassing the data processing inequality.

Step 1: Likelihood ratio in the WHT basis. By (C2), $f^{(r)}$ is a function of the DDT-derived distribution $p_r(\Delta C)$. Define the excess density

$$\ell_r(\delta) := p_r(\delta) - 2^{-b}.$$

In the WHT basis (cf. Proposition 1):

$$\ell_r(\delta) = 2^{-b} \sum_{u \neq 0} \hat{T}(u)^r \hat{p}_0(u) (-1)^{u \cdot \delta}.$$

Step 2: (C3) implies pointwise sign preservation of ℓ . We claim: for every δ ,

$$\text{sign}(\ell_r(\delta)) = \text{sign}(\ell_{r'}(\delta)) \quad \text{whenever both are nonzero.}$$

Write $\lambda_u := \hat{T}(u) \hat{p}_0(u)$ and $\mu_u(\delta) := \lambda_u (-1)^{u \cdot \delta}$, so $\ell_r(\delta) = 2^{-b} \sum_{u \neq 0} \hat{T}(u)^{r-1} \mu_u(\delta)$.

Condition (C3), via the WHT analysis of Proposition 1, requires $\hat{T}(u) \geq 0$ for all spectrally active u . Since $0 \leq \hat{T}(u) \leq 1$, the weights $\hat{T}(u)^{r-1}$ are nonneg and decrease in r . Therefore $\ell_r(\delta)$ and $\ell_{r'}(\delta)$ are nonneg-weighted sums of the *same* signed terms $\{\mu_u(\delta)\}$ —with the weights at r' dominating those at r componentwise.

A nonneg linear combination of real numbers that is positive remains positive when each weight is increased (while remaining nonneg). Hence the sign of the sum is preserved.

Step 3: From pointwise sign preservation to accuracy. Let $\mathcal{S}^+ := \{\delta : f^{(r)}(\delta) > \frac{1}{2}\}$ be the classifier’s “real” prediction region. Since $\text{acc}(f^{(r)}, \mathcal{D}_r) > \frac{1}{2}$, the classifier must satisfy

$$\sum_{\delta \in \mathcal{S}^+} (p_r(\delta) - 2^{-b}) > 0,$$

i.e., $\mathcal{S}^+$ is concentrated where $\ell_r > 0$.

By Step 2, $\text{sign}(\ell_r(\delta)) = \text{sign}(\ell_{r'}(\delta))$ for all δ . Therefore $\ell_{r'}(\delta) > 0$ on the same set, and

$$\text{acc}(f^{(r)}, \mathcal{D}_{r'}) - \frac{1}{2} = \sum_{\delta \in \mathcal{S}^+} (p_{r'}(\delta) - 2^{-b}) \geq 0.$$

Crucially, this argument does *not* require $f^{(r)}$ to be monotone in the biases, linear, or even continuous. It requires only that $f^{(r)}$ correctly classifies a majority of points under $\mathcal{D}_r$, and that (C3) preserves the pointwise sign of the likelihood ratio. Any classifier—linear, deep, or otherwise—satisfying these conditions is covered.

Remark on the DPI. The original proof sketch used the data processing inequality to argue $I(Y; \Delta C_r) \leq I(Y; \Delta C_{r'})$, which is correct but *insufficient*: high MI does not prevent a classifier from mapping features with the wrong polarity (this is precisely what the anti-transfer paradox demonstrates). Proposition 3 bypasses the DPI entirely, working instead with the pointwise likelihood ratio under (C3). The DPI remains useful for interpreting the MI paradox (Sect. 5.3) but is not the mechanism underlying positive transfer.